

LAB1

SRN:PES2UG24CS438

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs	High	Pass: Profile is published to showcase catalog upon validation of repository link. Fail: Invalid URL format accepted	Establishes the core discoverable record for each project; unvalidated links would break the showcase catalog's credibility.	
FR-002	Functional	The system shall allow Panel Judges to score submitted projects against a structured digital rubric (multiple weighted criteria, e.g., Innovation, Technical Depth, Presentation), with scores saved per judge per project.	High	Pass: Rubric score is saved only after all mandatory criteria fields are filled and the total is auto-calculated correctly. Fail: Submission accepted with missing rubric criteria or a miscalculated total.	Standardized, weighted scoring ensures fair and auditable evaluation across judges and projects.	
FR-003	Functional	The system shall allow members of the public to cast one vote per project per verified identity (e.g., OTP/email-verified account), preventing duplicate or bot-driven votes	High	Pass: A second vote attempt from the same verified identity on the same project is rejected with an error. Fail: The same identity is able to vote more than once on a single project.	Anti-sybil protection is essential to keep public voting results legitimate and trustworthy.	

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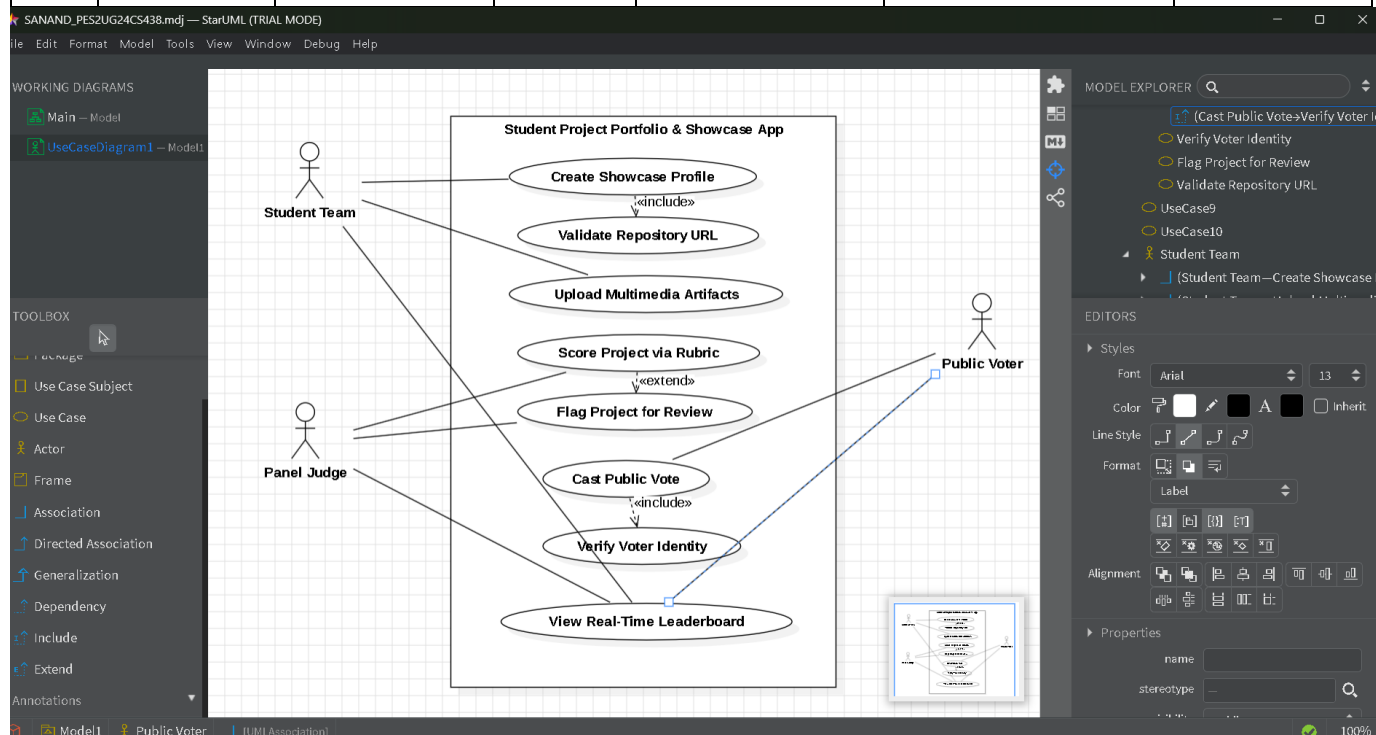
FR-004	Functional	The system shall generate and display a real-time public leaderboard ranking projects by a combined score (judge rubric score + normalized public vote count), recalculated on each new vote or score entry	Medium	Pass: Leaderboard position updates within the defined latency threshold immediately after a new vote/score is submitted. Fail: Leaderboard shows stale rankings after new votes/scores are recorded	A live leaderboard drives expo engagement and gives teams/judges/public immediate, transparent feedback.	
FR-005	Functional	The system shall allow Student Teams to upload and manage multimedia artifacts (posters, images, PDFs) attached to their showcase profile, subject to file type and size validation	Medium	Pass: Only files matching allowed types/size limits are accepted and attached to the profile. Fail: Oversized or disallowed file types are uploaded successfully. Rationale	Controlled multimedia uploads keep the showcase visually consistent and protect system storage/performance.	
NFR-001	Nonfunc.	The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.		
NFR-002	Nonfunc.	The system shall be accessible and fully functional on mobile browsers (responsive design) with 99.5% uptime during the official expo	High	Pass: UI renders correctly and all core actions (voting, scoring, profile viewing) complete successfully on mobile viewport widths (320px–768px); uptime monitoring	Voting and judging happen live, on-site, mostly via phones; downtime or broken mobile UX directly blocks time-critical, one-time stakeholder actions	

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		window, since judges and public voters primarily use handheld devices on-site.		confirms $\geq 99.5\%$ availability during the expo window. Fail: Layout breaks or core actions fail on mobile viewport, or downtime exceeds threshold during expo hours.		
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Use-Case Flow Specification

Use Case ID: UC6 **Use Case Name:** Cast Public Vote **Primary Actor:** Public Voter **Supporting Actors:** Identity Verification Service (external) **Related Use Cases:** «include» UC7 – Verify Voter Identity

Preconditions

1. The public voting module is open (expo voting window is active).
2. The project catalog is published and browsable.
3. The voter has access to the showcase platform (web/mobile) and a valid email or phone number for verification.

Postconditions

Success:

- Exactly one vote is recorded against the selected project, tied to the voter's verified identity.
- The project's vote count and combined leaderboard score are updated in real time.

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- The voter's identity is flagged internally as "voted for Project X" to block future duplicate votes.

Failure:

- No vote is recorded.
- The system state remains unchanged from before the attempt.

Main Success Scenario

1. The Public Voter browses the showcase catalog and selects a project to vote for.
2. The system displays the project profile with a "Vote" button.
3. The Public Voter clicks "Vote."
4. The system «includes» **UC7 – Verify Voter Identity**: a. The system prompts the voter to enter their email/phone number. b. The system sends a one-time password (OTP) via the Identity Verification Service. c. The voter enters the OTP. d. The system confirms the OTP is valid and matches an identity not already used to vote for this project.
5. The system records the vote against the project, linked to the verified identity.
6. The system recalculates the project's combined score (rubric score + normalized vote count).
7. The system updates the real-time public leaderboard.
8. The system displays a confirmation message: "Your vote has been recorded" to the voter.

Alternate Flow A1: Duplicate Vote Attempt (Anti-Sybil Rejection)

Trigger: At Step 4(d), the system detects the verified identity has already voted for this project.

1. The system halts the voting transaction.
2. The system displays an error message: "You have already voted for this project."
3. The system does **not** record a new vote and does **not** alter the existing vote count.
4. The system redirects the voter back to the project catalog.
5. Use case resumes at Step 1 (voter may browse and vote for a different, unvoted project) or ends.